

EXAM 7 – SPRING 2011

15. (2 points)

An insurance carrier has an internal model that determines its required economic capital.

a. (1 point)

Explain how the carrier's economic capital may not equal its:

- i. Rating agency capital.
- ii. Economic impairment earnings level.

b. (0.5 point)

Explain how the carrier's internal model might be changed in order to assess the economic impairment earnings level.

c. (0.5 point)

Briefly describe two situations where a single internal model may not be practical to use for multiple capital and ERM applications.

CONTINUED ON NEXT PAGE

Sample Solutions

Exam 07, 2011, Question 15

Solution 1

- a)
 - i. This may be calculated using a separate rating agency factor-based model.
 - ii. This “earning at risk” measurement is often assessed within an internal model at a lower return period so it may not equal to economic capital.
 - iii. In order to avoid reaching an “economic impairment earning level” a different value of capital may be required.
- b)

Economic impairment earnings level measurement is often assessed within an internal model at a lower return period (e.g. the 90% level of sufficiency).
- c)
 - i. Day-to-day business decisions (such as product approvals) are difficult to make with company level internal model that has a long run time.
 - ii. Groups may have a variety of models in use where the group is managed on decentralized lines.
 - iii. allocating capital back to lines of business may not be practical using just one model.
 - iv. carrier’s overall company-level model may be simpler than needed for product pricing.